For components of the workflow that can be directly implemented using existing command-line tools or libraries accessible via Python scripts (e.g., pandas, matplotlib), we ensured the following -

4.1 Existing command-line tools

Operations	Tool Uses	CLI Compatible	Action Taken
Reading CSV files	Pandas	Yes	Wrapped pandas.read_csv() into a CLI tool (read_csv.py)
Merging datasets by location	Pandas	Yes	Use pandas.merge() in merge_data.py
Testing correlations	Pandas	Yes	Use .corr() inside test_correlation.py
Comparing groups	Pandas	Yes	Use .groupby() in compare_groups.py
Visualizing results (plots)	Matplotlib	Yes	Use matplotlib in visualize.py to generate PNG plots

4.2 Missing tools

The following components required custom implementations due to project-specific logic or lack of existing CLI tools:

Operation	Tool Implementation	Descriptions
Treat missing values (-999 to NaN)	clean_missing.py	Converts placeholder values to NaN
Remove entries with missing location	check_location.py	Drops rows with null location
Save results to text format	to_text.py	Converts final dataframes into .txt summaries
Full pipeline test script	main_workflow.sh	Runs all components in sequence for reproducibilit